

Comprehensive Data Analytics Capstone Report

Cafe Retail Performance & Operational Intelligence

Internship Program: SWYNEX Technologies — Data Analytics Internship

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Project Title: End-to-End Retail Data Sanitation, Exploratory Analysis, and Executive Business Intelligence

Repository: [SWYNEX-Data-Analysis](#)

1. Executive Summary

This report delivers a full lifecycle data analytics evaluation of retail cafe point-of-sale (POS) operations based on 10,000 transaction records covering the full 2023 calendar year (dirty_cafe_sales.csv). The primary objective was to transform raw, defective transactional records into an enterprise-ready analytics asset, diagnose behavioral and financial trends, construct an interactive executive Power BI dashboard, and propose data-backed operational interventions.

Raw Uncurated Data (10,000 Dirty Rows) → Automated Pipeline (Python / Pandas) → Statistical Modeling (Pandas / Seaborn) → Power BI Dashboard (Executive BI) → Strategic Action (Food Bundling)

Through algorithmic sanitation, 100% of transaction records were preserved without arbitrary deletion. Exploratory modeling revealed a total annual revenue of \$89,304.50, an average order value (AOV) of \$8.93, and pronounced divergence between volume-generating and revenue-generating menu items.

2. Problem Statement & Operational Context

Point-of-sale architectures in high-throughput food and beverage establishments regularly suffer from upstream system anomalies, dropped payloads, and unstandardized manual entry. These flaws compromise business intelligence layers, distort tax/inventory reporting, and lead to poor procurement planning.

The primary business hurdles identified in the source system include:

- Corrupted quantitative measurements blocking mathematical aggregations.
- Incomplete categorizations obscuring customer purchase preferences and fulfillment channel splits.
- Lack of real-time visibility into product performance, seasonal demand variance, and basket sizing.

3. Dataset Architecture & Defect Diagnosis

The initial dataset (dirty_cafe_sales.csv) comprised 10,000 records across 8 core operational dimensions. A rigorous pre-cleaning data audit identified systematic data defects across all fields:

Field Name	Raw Type	Corrupted / Missing Entries	Impact on Analysis
Transaction ID	object	0	Unique key identifier; intact.
Item	object	969 rows (333 nulls + 636 corrupted strings)	Obscures product-level contribution.
Quantity	object	479 rows (138 nulls + 341 corrupted strings)	Blocks volume metrics and basket size distributions.
Price Per Unit	object	533 rows (179 nulls + 354 corrupted strings)	Prevents margin calculations and price elasticity checks.
Total Spent	object	502 rows (173 nulls + 329 corrupted strings)	Directly distorts gross financial accounting.
Payment Method	object	3,178 rows (2,579 nulls + 599 corrupted strings)	High sparsity (~31.8%); blinds payment gateway optimization.
Location	object	3,961 rows (3,265 nulls + 696 corrupted strings)	High sparsity (~39.6%); impairs store vs. takeaway logistics.
Transaction Date	object	460 rows (159 nulls + 301 corrupted strings)	Disables time-series analysis and seasonality tracking.

4. Data Engineering & Sanitization Pipeline

To remediate these issues deterministically, an automated Python/Pandas cleaning script was engineered:

Raw Input → Sentinel Mapping → Deterministic Imputation → Arithmetic Identity → Schema Enforcement

4.1 Sentinel Token Standardization

Raw placeholder text strings ("ERROR", "UNKNOWN") were systematically transformed into standardized IEEE NaN floats, allowing Pandas vectorized numerical functions to execute cleanly without syntax aborts.

4.2 Deterministic Menu Matrix Imputation

Rather than applying generic mean or median values (which corrupt variance and introduce decimal pricing), a deterministic menu price matrix was mined from uncorrupted transactions:

Menu Item	Item Price Reference
Salad	\$5.00
Sandwich	\$4.00
Smoothie	\$4.00
Cake	\$3.00

Menu Item	Item Price Reference
Juice	\$3.00
Coffee	\$2.00
Tea	\$1.50
Cookie	\$1.00

Missing unit prices were imputed by mapping against this known price vector.

4.3 Mathematical Constraint Reconciliation

Where either Total Spent or Quantity was unpopulated, the missing metric was recomputed via fundamental transactional constraints:

$$\text{Total Spent} = \text{Quantity} \times \text{Price Per Unit}$$

$$\text{Quantity} = \lfloor \text{Total Spent} \div \text{Price Per Unit} \rfloor$$

4.4 Categorical & Temporal Typings

High-sparsity categorical missing values in Location and Payment Method were filled with an explicit "Unknown" classification to preserve sample size while isolating unrecorded entries. Date fields were parsed into standardized ISO datetime64[ns] timestamps.

5. Statistical Exploratory Data Analysis & Business Insights

Analysis of the validated 10,000-record dataset (cleaned_cafe_sales.csv) produced concrete empirical benchmarks:

5.1 Macro Performance Metrics

- **Total Gross Revenue:** \$89,304.50
- **Total Transactions:** 10,000
- **Gross Unit Volume Sold:** 30,249 units
- **Mean Basket Value (AOV):** \$8.93 ($\sigma = \6.00)
- **Median Spend:** \$8.00
- **Order Size Range:** 1 to 5 units ($\mu = 3.02$, Median = 3.00)

5.2 Menu Performance & Contribution Analysis

Menu Item	Price Per Unit	Units Sold	Total Revenue	Share of Total Spend	Mean Ticket
Salad	\$5.00	3,475	\$17,375.00	19.46%	\$15.14
Sandwich	\$4.00	3,441	\$13,764.00	15.41%	\$12.17
Smoothie	\$4.00	3,342	\$13,368.00	14.97%	\$12.20
Juice	\$3.00	3,514	\$10,542.00	11.80%	\$9.00

Menu Item	Price Per Unit	Units Sold	Total Revenue	Share of Total Spend	Mean Ticket
Cake	\$3.00	3,471	\$10,413.00	11.66%	\$9.14
Coffee	\$2.00	3,551	\$7,102.00	7.95%	\$6.10
Tea	\$1.50	3,310	\$4,965.00	5.56%	\$4.56
Cookie	\$1.00	3,238	\$3,238.00	3.63%	\$2.97
Unknown	Variable	2,907	\$8,537.50	9.56%	\$8.81

Revenue by Menu Item

Menu Item	Relative Revenue	Revenue
Salad		\$17.4K
Sandwich		\$13.8K
Smoothie		\$13.4K
Juice		\$10.5K
Cake		\$10.4K
Coffee		\$7.1K
Tea		\$5K
Cookie		\$3.2K

5.3 Detailed Analytical Findings

- **Salad Serves as the Revenue Anchor:** Salad produced the highest cumulative earnings (\$17,375.00) and elevated ticket values (\$15.14) due to its \$5.00 price point, outperforming core cafe beverages by more than 2:1.
- **The High-Volume Beverage Disconnect:** Coffee generated the greatest raw units ordered across the entire menu (3,551 units across 1,165 tickets) but yielded only \$7,102.00 in top-line revenue. It functions primarily as a customer acquisition draw rather than a direct basket builder.
- **Mid-Year Revenue Peaking:** Sales demonstrated seasonal consistency around a \$7,442.00 monthly baseline, reaching an annual maximum in June (\$7,799.50) and a localized minimum in February (\$7,032.00).
- **Uniform Order Distribution:** Quantities ordered across tickets were distributed evenly between 1 and 5 units (~2,000 orders each), confirming balanced demand between solo takeaway customers and multi-item group orders.
- **Payment Method Equidistribution:** Known payment channels displayed near-equal share: Digital Wallets (33.6%), Credit Cards (33.3%), and Cash (33.1%).

6. Business Intelligence Dashboard Design (Power BI)

To support executive decision-making, an interactive dashboard was built in Microsoft Power BI Desktop featuring an espresso-and-cream aesthetic (#FDF8F2 background, #4E342E espresso primary bars, #8D6E63 mocha column bars, and #FFFFFF cards).

Dashboard Layout: Cafe Sales Executive Dashboard — SWYNEX Technologies

TOTAL REVENUE	TOTAL ORDERS	UNITS SOLD	AVERAGE ORDER VALUE
\$89,304.50	10,000	30,249	\$8.93

Component	Content
Filters & Slicers	Location; Payment Method; Menu Item Filter; Date Timeline
Bar Chart	Total Revenue by Menu Item (\$k): Salad \$17.4k Sandwich \$13.8k Smoothie \$13.4k
Donut Chart	Payment Method Share: Wallet 33.6% Card 33.3% Cash 33.1%
Column Chart	Quantity, 1 to 5 items (~2k each)

Dashboard Core Functionalities:

- **Real-time KPI Tiles:** Immediate operational tracking of Total Revenue, Order Volume, Units Sold, and AOV.
- **Synchronized Slicers:** Dynamic cross-filtering by Store Location (In-store, Takeaway), Menu Item, Payment Method, and Date Slider.
- **Bi-directional Visual Interaction:** Selecting any visual segment updates all corresponding visual components across the page in real time.

7. Strategic Recommendations & Action Plan

Strategic Initiative	Identified Bottleneck	Proposed Intervention	Projected Operational Impact
Morning F&B Pairing	Coffee has high order frequency (3,551 units) but low average ticket value (\$6.10).	Bundle morning coffee with bakery or breakfast sandwiches (\$5.50 combo).	Lifts baseline beverage tickets toward the \$8.93 storewide average.
Digital Menu Promotion	Salad represents the highest revenue driver (\$17,375.00) and ticket size (\$15.14).	Position salads as primary combo upgrades on digital order boards.	Amplifies high-margin lunch ticket values.
Q1 Seasonality Mitigation	Annual revenue hits its lowest point in February (\$7,032.00).	Launch targeted seasonal loyalty promotions and limited-time warm drink offerings.	Flattens annual revenue volatility across Q1.
Omnichannel POS Uptime	Payment channels are split equally between digital wallets, cards, and cash.	Maintain dual redundancy across card/contactless terminals while preserving cash stations.	Prevents checkout bottlenecks and minimizes transaction abandonment.

8. Conclusion & Technical Verification

This project proves that deterministic, rule-based data cleaning pipelines can recover complete transaction logs without sacrificing sample volume or introducing statistical bias. The combination of Python-based exploratory modeling and Power BI visual storytelling delivers a transparent, end-to-end analytics foundation ready for direct enterprise deployment.